

# Seminar 6: Correlation and WLLN

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In many applications we are interested not only in the behavior of a single random variable, but also in the relationship between two random variables.

Covariance measures the direction of the linear relationship between two variables, while the correlation coefficient gives a normalized measure that always lies between  $-1$  and  $1$ .

These quantities are fundamental throughout statistics, machine learning, finance, econometrics, and data analysis.

## 1 Covariance

Let  $\xi$  and  $\eta$  be two random variables with finite expectations.

The **covariance** between  $\xi$  and  $\eta$  is defined as

$$\text{Cov}(\xi, \eta) = \mathbb{E}[(\xi - \mathbb{E}\xi)(\eta - \mathbb{E}\eta)].$$

Using linearity of expectation,

$$\boxed{\text{Cov}(\xi, \eta) = \mathbb{E}[\xi\eta] - \mathbb{E}[\xi] \mathbb{E}[\eta].}$$

Intuitively,

- positive covariance means that both variables tend to increase together;
- negative covariance means that one variable tends to increase while the other decreases;
- covariance close to zero indicates the absence of a linear relationship (although nonlinear dependence may still exist).

## 1.1 Properties of Covariance

For any random variables  $\xi, \eta, \xi_1, \xi_2$  and constants  $a, b$ ,

### 1. Symmetry

$$\text{Cov}(\xi, \eta) = \text{Cov}(\eta, \xi).$$

### 2. Covariance with itself

$$\text{Cov}(\xi, \xi) = \text{Var}(\xi).$$

### 3. Linearity

$$\text{Cov}(a\xi_1 + b\xi_2, \eta) = a \text{Cov}(\xi_1, \eta) + b \text{Cov}(\xi_2, \eta).$$

### 4. Independence implies zero covariance

If  $\xi \perp\!\!\!\perp \eta$ , then

$$\text{Cov}(\xi, \eta) = 0.$$

However, the converse is generally false: zero covariance does **not** imply independence.

## 1.2 Variance of a Sum

One of the most useful applications of covariance is computing variances of sums.

For any random variables  $\xi_1, \dots, \xi_n$ ,

$$\text{Var}\left(\sum_{i=1}^n \xi_i\right) = \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(\xi_i, \xi_j).$$

Equivalently,

$$\text{Var}\left(\sum_{i=1}^n \xi_i\right) = \sum_{i=1}^n \text{Var}(\xi_i) + 2 \sum_{i < j} \text{Cov}(\xi_i, \xi_j).$$

If the variables are independent, all covariance terms vanish, giving

$$\text{Var}\left(\sum_{i=1}^n \xi_i\right) = \sum_{i=1}^n \text{Var}(\xi_i).$$

## 2 Correlation Coefficient

Since covariance depends on the units of measurement, it is difficult to compare across different datasets.

The **correlation coefficient** normalizes covariance by the standard deviations:

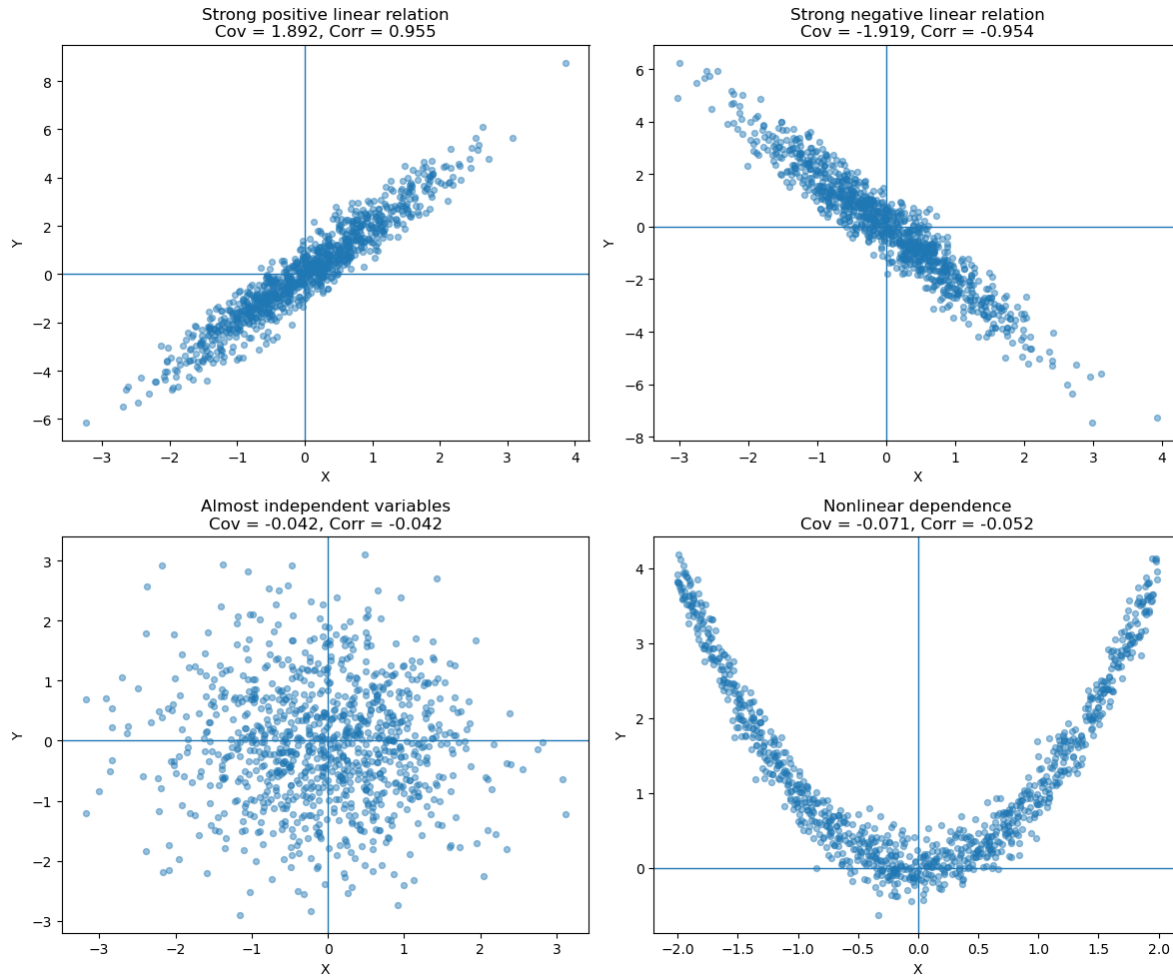
$$\text{Corr}(\xi, \eta) = \frac{\text{Cov}(\xi, \eta)}{\sqrt{\text{Var}(\xi)} \sqrt{\text{Var}(\eta)}}.$$

The correlation coefficient is dimensionless and satisfies

$$-1 \leq \text{Corr}(\xi, \eta) \leq 1.$$

Its interpretation is straightforward:

- $\text{Corr} = 1$  : perfect positive linear relationship;
- $\text{Corr} = -1$  : perfect negative linear relationship;
- $\text{Corr} = 0$  : no linear relationship.



## 2.1 Example: GDP Growth and Association Between Two Countries

The following data (this is not real data BTW) represent GDP growth indicators (in percent relative to the year 2005) for **Russia** and **Belarus** over the same 14-year period.

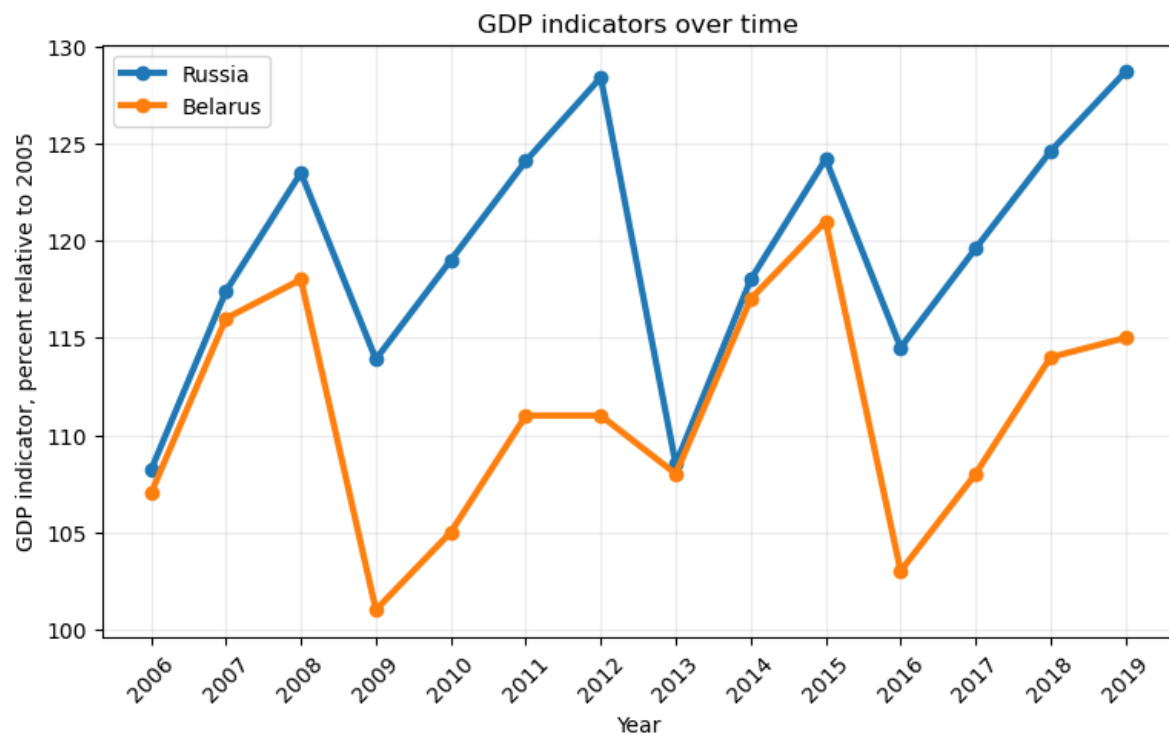
**Russia**

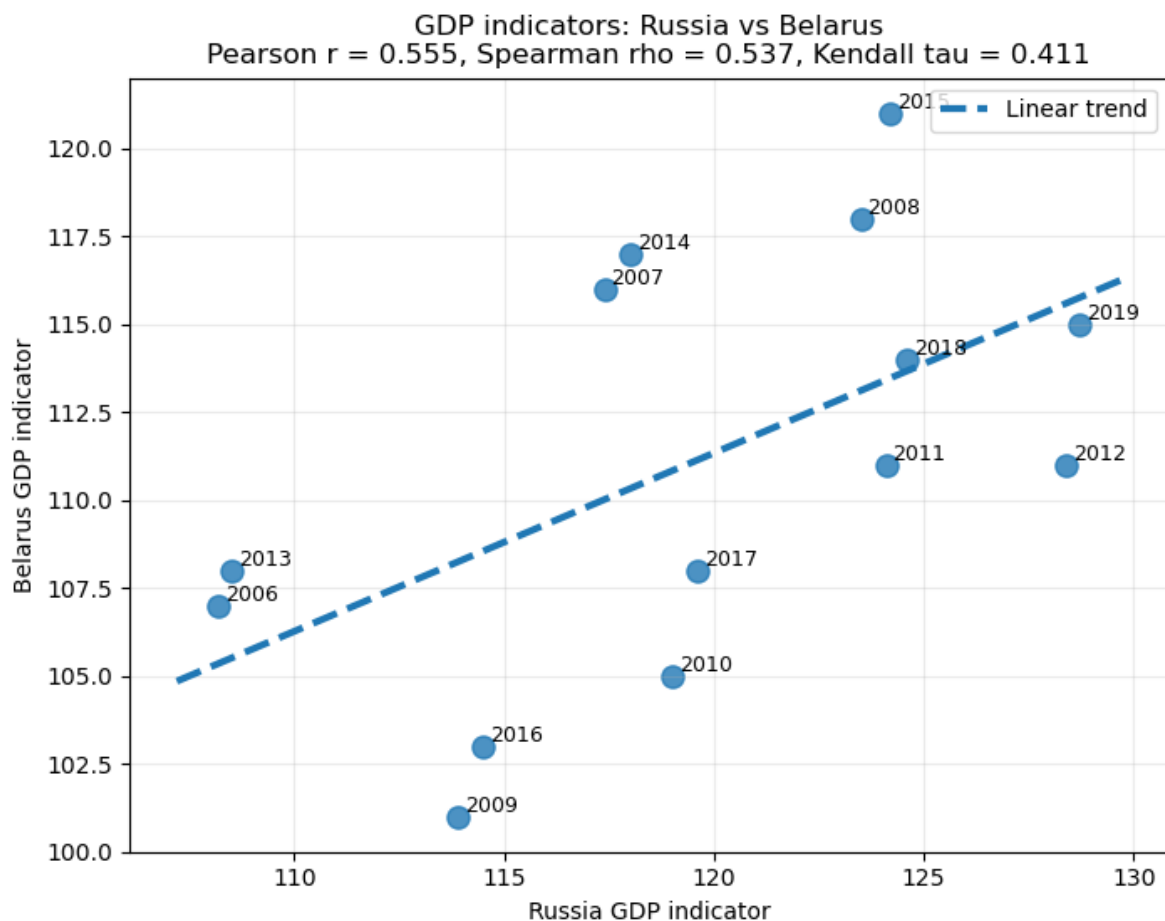
108.2, 117.4, 123.5, 113.9, 119.0, 124.1, 128.4, 108.5, 118.0, 124.2, 114.5, 119.6, 124.6, 128.7

**Belarus**

107, 116, 118, 101, 105, 111, 111, 108, 117, 121, 103, 108, 114, 115

Our goal is to determine whether these two samples appear to be statistically independent.





### 3 Concentration Inequalities

In many situations the exact distribution of a random variable is unknown or difficult to compute.

Instead of finding exact probabilities, we seek **upper bounds** for probabilities of large deviations from typical values.

#### 3.1 Markov's Inequality

Let  $\xi \geq 0$  be a nonnegative random variable with finite expectation.

Then for every  $a > 0$ ,

$$\mathbb{P}(\xi \geq a) \leq \frac{\mathbb{E}[\xi]}{a}.$$

This inequality requires only the existence of the expectation and makes no assumptions on the distribution. Although often crude, it serves as the starting point for many stronger inequalities.

### 3.2 Chebyshev's Inequality

Suppose  $\xi$  has finite expectation and finite variance.

Then for every  $\varepsilon > 0$ ,

$$\mathbb{P}(|\xi - \mathbb{E}\xi| \geq \varepsilon) \leq \frac{\text{Var}(\xi)}{\varepsilon^2}.$$

Unlike Markov's inequality, Chebyshev's inequality uses both the expectation and the variance.

## 4 Weak Law of Large Numbers

Suppose  $X_1, X_2, \dots$  are independent and identically distributed random variables satisfying

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty.$$

The sample mean is  $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ .

The Weak Law of Large Numbers states that

$$\bar{X}_n \xrightarrow{P} \mu.$$

In other words,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) = 0 \quad \text{for every } \varepsilon > 0.$$

## 4.1 Bernoulli Trials and the Interpretation of Probability

One of the most important consequences of the Weak Law of Large Numbers is that it provides a mathematical justification for interpreting **probability as long-run relative frequency**.

Suppose we repeat the same experiment independently many times. Let

$$X_i = \begin{cases} 1, & \text{if the event occurs on trial } i, \\ 0, & \text{otherwise.} \end{cases}$$

Then  $X_i \sim \text{Bernoulli}(p)$ . Therefore

$$\mathbb{E}[X_i] = p, \quad \text{Var}(X_i) = p(1 - p).$$

Notice that  $\sum_{i=1}^n X_i$  is exactly the number of times the event occurred during the first  $n$  trials. Therefore,

$$\bar{X}_n = \frac{\text{Number of successes}}{\text{Number of trials}},$$

which is precisely the **empirical frequency** (or relative frequency) of the event.

Applying the Weak Law of Large Numbers,

$$\bar{X}_n \xrightarrow{P} p,$$

or equivalently,

$$\boxed{\lim_{n \rightarrow \infty} \mathbb{P} \left( \left| \frac{\text{Number of successes}}{n} - p \right| > \varepsilon \right) = 0 \quad (\varepsilon > 0).}$$

Thus, as the number of independent repetitions increases, the observed frequency of the event becomes arbitrarily close to its true probability with probability approaching one.

This result gives a rigorous mathematical foundation for the frequentist interpretation of probability:

**The probability of an event is the limiting value of its relative frequency in a large number of independent repetitions of the same experiment.**



